

Questions with Answer Keys

MathonGo

Q1: 16 March (Shift 1) - Numerical

A fringe width of 6 mm was produced for two slits separated by 1 mm apart. The screen is placed 10 m away. The wavelength of light used is 'x' nm. The value of 'x' to the nearest integer is _____

Q2: 18 March (Shift 1) - Single Correct

In Young's double slit arrangement, slits are separated by a gap of 0.5 mm, and the screen is placed at a distance of 0.5 m from them. The distance between the first and the third bright fringe formed when the slits are illuminated by a monochromatic light of 5890 Å is :-

(1) $1178 \times 10^{-9} \text{ m}$

(2) $1178 \times 10^{-6} \text{ m}$

(3) $1178 \times 10^{-12} \text{ m}$

(4) $5890 \times 10^{-7} \text{ m}$

Q3: 18 March (Shift 2) - Single Correct

The speed of electrons in a scanning electron microscope is $1 \times 10^7 \text{ ms}^{-1}$. If the protons having the same speed are used instead of electrons, then the resolving power of scanning proton microscope will be changed by a factor of:

(1) 1837

(2) $\frac{1}{1837}$

(3) $\sqrt{1837}$

(4) $\frac{1}{\sqrt{1837}}$

Answer Key

Q1 (600)

Q2 (2)

Q3 (1)

Q1 (600)

$$\beta = \frac{\lambda D}{d}$$

$$\lambda = \frac{\beta d}{D}$$

$$\lambda = \frac{6 \times 10^{-3} \times 10^{-3}}{10}$$

$$\lambda = 6 \times 10^{-7} \text{ m} = 600 \times 10^{-9} \text{ m}$$

$$\lambda = 600 \text{ nm}$$

Q2 (2)

$$\beta = \frac{\lambda D}{d} = \frac{5890 \times 10^{-10} \times 0.5}{0.5 \times 10^{-3}}$$

$$= 589 \times 10^{-6} \text{ m}$$

Distance between first and third bright fringe is $2\beta = 2 \times 589 \times 10^{-6} \text{ m}$

$$= 1178 \times 10^{-6} \text{ m}$$

Ans.(b)

Q3 (1)

$$\text{Resolving power (RP)} \propto \frac{1}{\lambda}$$

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

$$\text{So (RP)} \propto \frac{mv}{h}$$

$$\text{RP} \propto p$$

$$\text{RP} \propto mv$$

$$\text{RP} \propto m$$